

Mohamed Elnaggar

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Education

University of Calgary – B.Sc. in Software Engineering

June 2025

Schulich's Engineering Design Fair Silver Medal | UCalgary President's UG Award

Experience

Co-Founder & Full Stack Engineer, Zaymar Creatives – Fresno, CA (Remote)

Oct 2025 – Present

- Co-founded the company and served as the sole engineer across all client systems, independently taking care of every architecture, implementation, and deployment decision.
- Delivered four production platforms end-to-end, standardizing CI/CD pipelines using GitHub Actions and Vercel to enable automated builds, testing, and deployment across projects.
- Applied systematic quality practices across all systems: proactive CVE auditing, input validation and error boundaries on user-facing workflows, graceful recovery under network failures, and automated regression checks via Vercel CI/CD.
- Built Node.js ETL pipelines to automate unstructured dataset processing, reducing manual data handling by 90%.

Software Telemetry Member, UCalgary Racing – Calgary, AB

Sept 2024 – May 2025

- Developed and validated CAN bus telemetry ingestion pipelines by simulating real-time data streams, ensuring consistent data flow across distributed components.
- Evaluated MQTT, WebSockets, Socket.io, Raw UDP, and Raw TCP using a dual-clock methodology, measuring wall-clock inter-packet delay against CAN-timestamp-derived expected delay simultaneously for ground-truth comparison.
- Built a worker pool tuned to Raspberry Pi core count, a 1,000-deep buffered job channel, sync.pool for byte-slice reuse to reduce GC pressure, and non-blocking channel sends with explicit backpressure handling.
- Three-server design (raw telemetry receiver, REST historical API, live broadcast), each with graceful shutdown sequencing, with a priority fast-path for cell voltage frames decoded inline, bypassing the worker pool entirely.
- Conducted load, stress, and spike testing to evaluate system availability and fault tolerance under real-world conditions.
- Benchmarked Native PostgreSQL vs. TimescaleDB at thousands of reads and writes/sec. Chose TimescaleDB for its superior time-series query performance despite native PostgreSQL's write speed advantage on the Pi's ARM architecture.

Projects

Attune (In Development) - React Native, Expo, Convex, Clerk, Zustand

- Focus and performance platform for university students targeting digital distraction and academic performance anxiety.
- Implemented AppState-resilient session timing with zero time-drift across iOS background states, hardware-clock distraction logging, and a data-driven session recap with focus scores and XP rewards.
- AI-powered personalization layer planned following a structured literature review of CBT and focus research.

FSAE Real-Time Telemetry System - Go, Python, PostgreSQL, Bash, Docker, React

- Designed a hybrid event-driven and microservices architecture for real-time telemetry processing, validated through early-stage prototyping.
- Implemented real-time communication using WebSockets, achieving 110 ms latency for sensor-to-server data transfer.
- Containerized the backend and web client with Docker, ensuring seamless deployment with production Bash setup scripts featuring architecture detection, pinned dependency versions, and DB connectivity pre-validation.
- The team placed 2nd and 3rd across multiple categories at the 2026 Michigan FSAE Hybrid competition.

Fresno Auto Sales - Next.js, Sanity CMS, React, Tailwind, Framer Motion

- Prioritized a modern UI with an accessibility-conscious layout, responsive design across all device sizes, and Core Web Vitals performance to ensure a consistent experience across a diverse user base, minimizing friction across the full user journey.
- Enforced data consistency using Sanity's structured content model to prevent malformed states from reaching production.

Skills

Languages & Dev: Go, Python, TypeScript/JavaScript, Java, C/C++, Bash, Node.js, Next.js, ETL Pipelines

Systems & Infrastructure: System Design, REST APIs, Performance Testing, Docker, AWS, CI/CD (GitHub Actions), Cloudflare, Git

Data & ML: Scikit-learn, TensorFlow, Pandas, NumPy, PostgreSQL, TimescaleDB, SQL, Supabase, Convex, Data Validation & Modeling

Research & Testing: Qualitative studies, Multi-protocol benchmarking, dual-clock latency analysis, load/stress/spike testing